# MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

## OVERALL SEVERITY: CRITICAL

| | |
|---|---|
| **Case ID** | `case01_baseline_vs_case02_t1055_5_attack` |
| **Generated** | `2026-04-19 19:10:53 UTC` |
| **Baseline capture** | `case01_baseline` |
| **Attack capture** | `case02_t1055_5_attack` |
| **Baseline path** | `/MFF/cases/case01_baseline` |
| **Attack path** | `/MFF/cases/case02_t1055_5_attack` |

# 1. EXECUTIVE SUMMARY

| Metric | Value | Metric | Value |
|---|---|---|---|
| New Processes (Attack Only) | 13 | Critical Risk Processes | 3 |
| Gone Processes (Baseline Only) | 9 | ATT&CK Techniques Observed | 2 |
| New Network Connections | 131 | ATT&CK Tactics Observed | 1 |
| Flagged Connections | 3 | IOCs Extracted | 40 |

### ATT&CK; Tactics Observed:

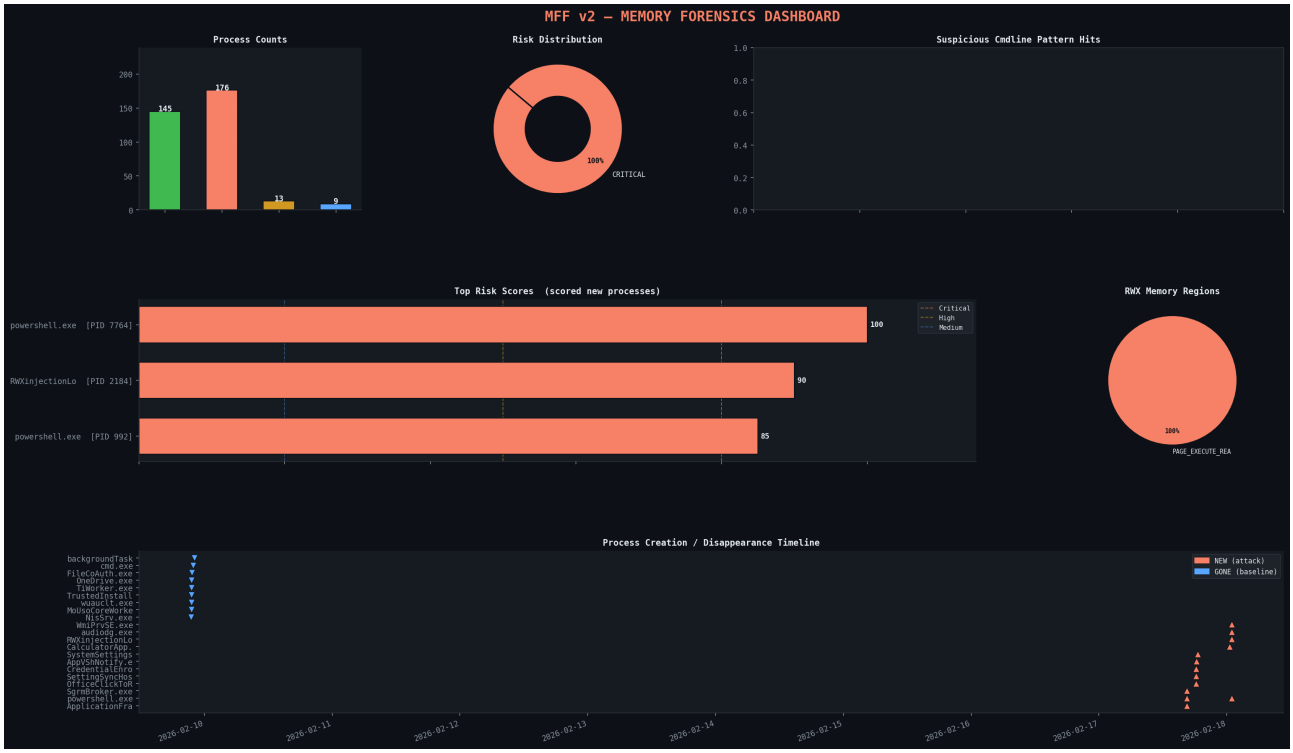
Defense Evasion

### IOC Summary:

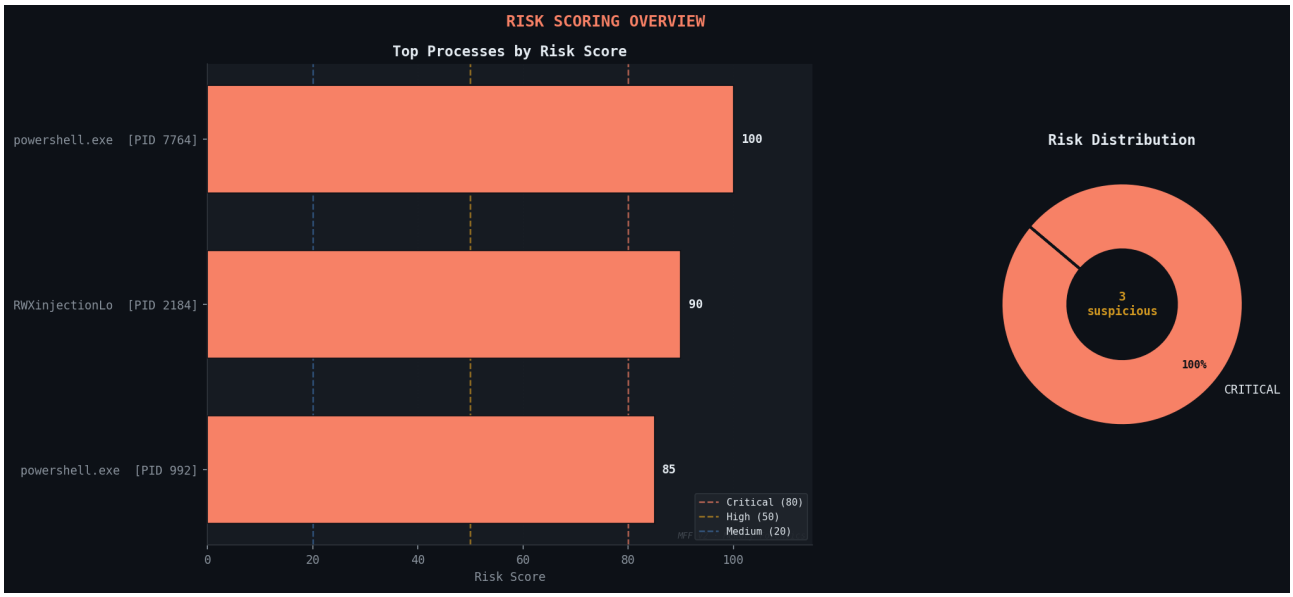
| Type | Indicators (top 8) |
|---|---|
| IPv4 | 0.0.0.0 · 150.171.28.11 |

# 2. VISUALIZATIONS

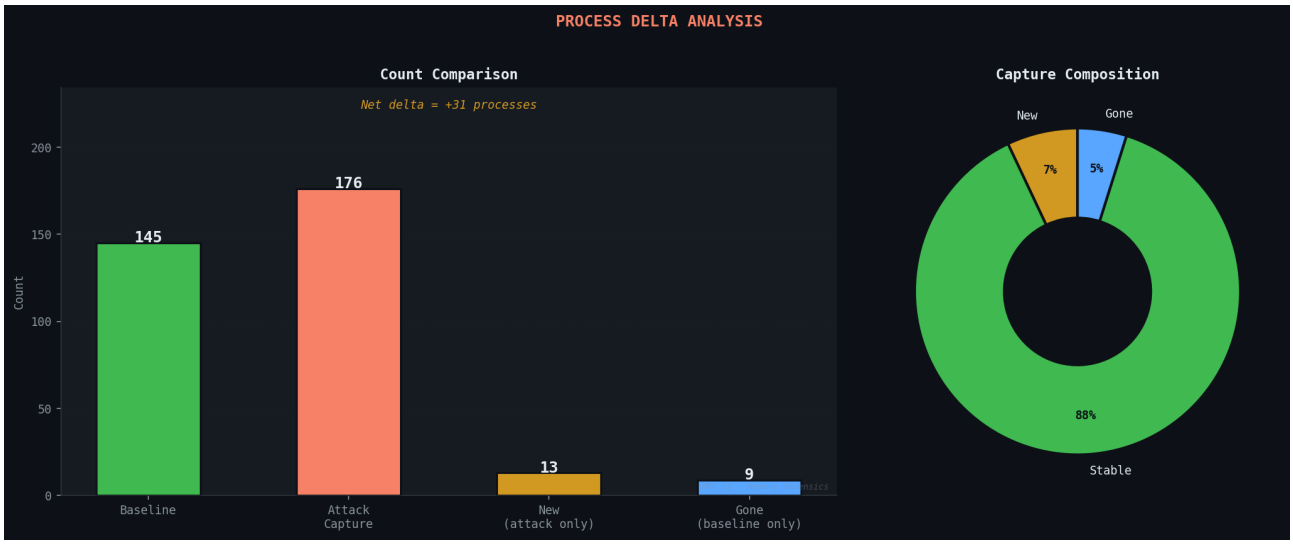
## 2.1 Full Forensics Dashboard



## 2.2 Risk Scoring Overview

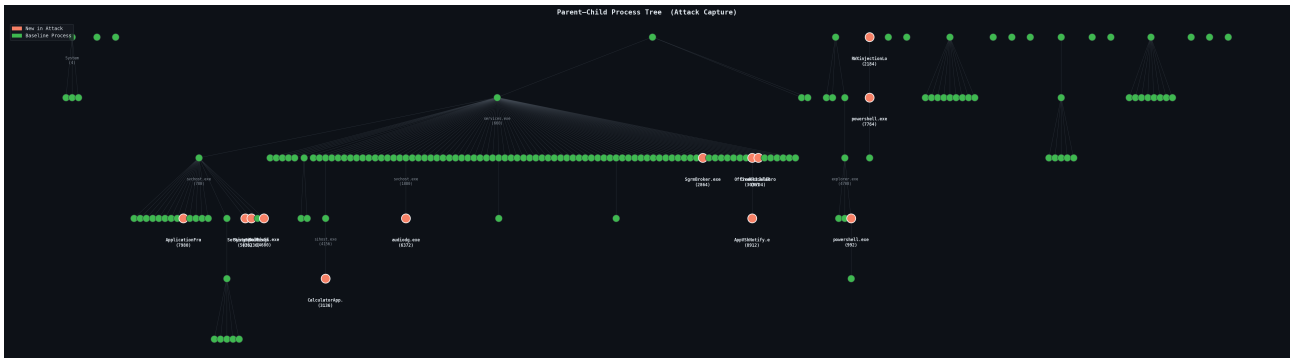


## 2.3 Process Delta Analysis

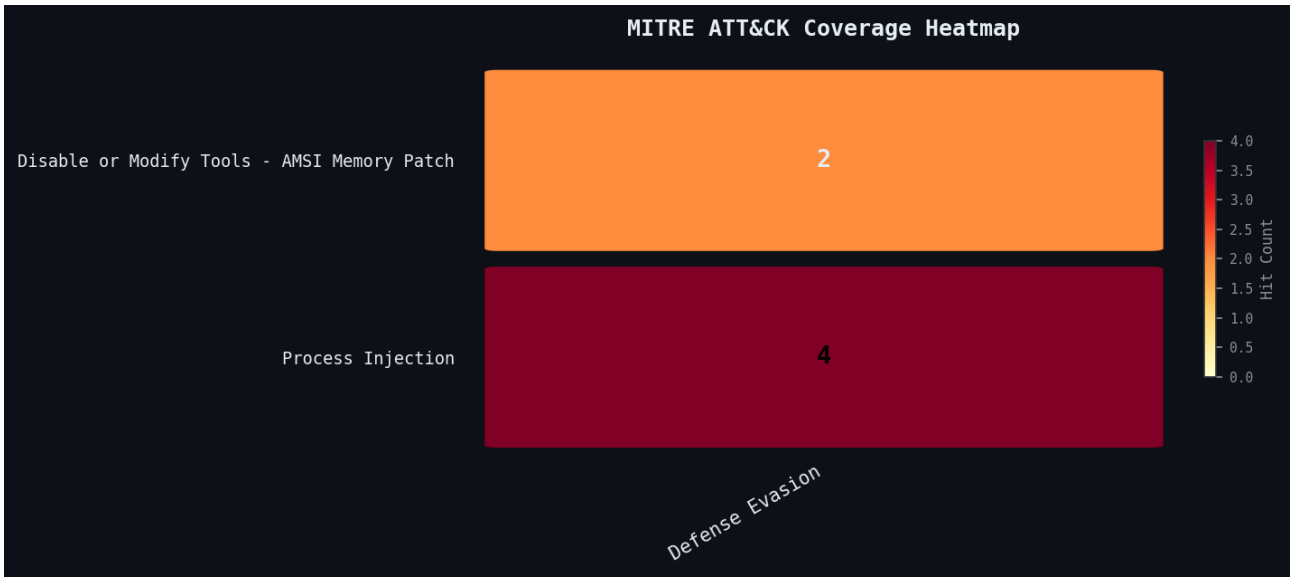



## 2.4 Process Timeline

## 2.5 Parent-Child Process Tree



## 2.6 MITRE ATT&CK; Heatmap




## 2.8 Malfind Memory Protections

## 3. RISK SCORE TABLE

| Process | PID | Score | Level | Indicators | | |
|---|---|---|---|---|---|---|
| powershell.exe | 7764 | 100 | CRITICAL | T1562.001 | 3 src | High |
| RWXinjectionLo | 2184 | 90 | CRITICAL | T1055 | 1 src | Medium |
| powershell.exe | 992 | 85 | CRITICAL | T1562.001 | 2 src | Medium-High |

## 4. NEW PROCESSES (Attack Only)

| ImageFileName | PID | PPID | CreateTime |
|---|---|---|---|
| ApplicationFra | 7980 | 780 | 2026-02-17 16:30:03.000000 UTC |
| powershell.exe | 992 | 4708 | 2026-02-17 16:30:29.000000 UTC |
| SgrmBroker.exe | 2864 | 660 | 2026-02-17 16:30:58.000000 UTC |
| OfficeClickToR | 3016 | 660 | 2026-02-17 18:12:01.000000 UTC |
| SettingSyncHos | 5076 | 780 | 2026-02-17 18:14:10.000000 UTC |
| CredentialEnro | 9704 | 660 | 2026-02-17 18:14:21.000000 UTC |
| AppVShNotify.e | 8912 | 3016 | 2026-02-17 18:18:11.000000 UTC |
| SystemSettings | 10236 | 780 | 2026-02-17 18:32:00.000000 UTC |
| CalculatorApp. | 3136 | 4156 | 2026-02-18 00:31:14.000000 UTC |
| RWXinjectionLo | 2184 | 9076 | 2026-02-18 00:55:10.000000 UTC |
| powershell.exe | 7764 | 2184 | 2026-02-18 00:55:10.000000 UTC |
| audiodg.exe | 6372 | 1880 | 2026-02-18 00:55:11.000000 UTC |
| WmiPrvSE.exe | 4680 | 780 | 2026-02-18 00:55:16.000000 UTC |

## 5. MITRE ATT&CK; TECHNIQUE MATCHES

| Technique | TechniqueName | Tactic | MatchedKeyword | PID | Process |
|---|---|---|---|---|---|
| T1055 | Process Injection | Defense Evasion | PAGE_EXECUTE_READWRITE | 992 | powershell.exe |
| T1055 | Process Injection | Defense Evasion | PAGE_EXECUTE_READWRITE | 7764 | powershell.exe |
| T1055 | Process Injection | Defense Evasion | EXECUTE_READWRITE | 992 | powershell.exe |
| T1055 | Process Injection | Defense Evasion | EXECUTE_READWRITE | 7764 | powershell.exe |
| T1562.001 | Disable or Modify Tools - AMSI Memory Patch | Defense Evasion | amsi.dll | 992 | powershell.exe |
| T1562.001 | Disable or Modify Tools - AMSI Memory Patch | Defense Evasion | amsi.dll | 7764 | powershell.exe |

## 6. SUSPICIOUS CMDLINE FINDINGS

No suspicious command lines found.

## 7. MALFIND — MEMORY INJECTION ARTEFACTS

| Process | PID | Protection |
|---|---|---|
| powershell.exe | 992 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 992 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7764 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7764 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7764 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7764 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7764 | PAGE_EXECUTE_READWRITE |

## 7b. DLL ANALYSIS — T1574 / T1562 FINDINGS

2 DLL finding(s) — T1574 DLL Search Order Hijacking + T1562 AMSI Bypass evidence.

| PID | Process | DLL | HijackType | Technique | RiskScore | Tactic |
|---|---|---|---|---|---|---|
| 992 | powershell.exe | amsi.dll | AMSI_BYPASS_MEMORY_PATCH | T1562.001 | 80 | Defense Evasion |
| 7764 | powershell.exe | amsi.dll | AMSI_BYPASS_MEMORY_PATCH | T1562.001 | 80 | Defense Evasion |

### [T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=992)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=992) loads amsi.dll AND has 2 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

### [T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=7764)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=7764) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

## 8. NETWORK ARTEFACTS

### 8a. Flagged Connections (suspicious ports)

| TreeDepth | Offset | Proto | LocalAddr | LocalPort | ForeignAddr | ForeignPort | State | PID | Owner | Created | ImageFileName | SuspiciousPort | PortMeaning |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0xdb84c157223 0 | TCPv4 | 0.0.0.0 | 135 | 0.0.0.0 | 0 | LISTENING | 900 | svchost.exe | 2026-02-17 15:28:51.000000 UTC | svchost.exe | 135 | DCOM/RPC |
| 0 | 0xdb84c157223 0 | TCPv6 | :: | 135 | :: | 0 | LISTENING | 900 | svchost.exe | 2026-02-17 15:28:51.000000 UTC | svchost.exe | 135 | DCOM/RPC |
| 0 | 0xdb84c1572e90 | TCPv4 | 0.0.0.0 | 135 | 0.0.0.0 | 0 | LISTENING | 900 | svchost.exe | 2026-02-17 15:28:51.000000 UTC | svchost.exe | 135 | DCOM/RPC |

## 9. IOC DETAIL TABLE

| Type | Value | Source | Count |
|---|---|---|---|
| FilePath | C:\\Windows\\system32\\svchost.exe | cmdline | 46 |
| FilePath | C:\\Windows\\System32\\svchost.exe | cmdline | 29 |
| FilePath | C:\\Program | cmdline | 9 |
| FilePath | C:\\Windows\\System32\\RuntimeBroker.exe | cmdline | 6 |
| FilePath | C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView | cmdline | 4 |
| FilePath | Microsoft\\EdgeWebView\\Application\\144.0.3719.115\\msedgewebview2.exe | cmdline | 4 |
| FilePath | ??\\C:\\Windows\\system32\\conhost.exe | cmdline | 2 |
| FilePath | C:\\ProgramData\\Microsoft\\Windows | cmdline | 2 |
| FilePath | C:\\Windows\\system32\\DllHost.exe | cmdline | 2 |
| FilePath | Microsoft\\Edge\\Application\\msedge.exe | cmdline | 2 |
| FilePath | Windows | cmdline | 2 |
| FilePath | system32\\csrss.exe | cmdline | 2 |
| FilePath | ??\\C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe | cmdline | 1 |
| FilePath | C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad | cmdline | 1 |
| FilePath | C:\\Windows\\Explorer.EXE | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\VBoxService.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\VBoxTray.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\spoolsv.exe | cmdline | 1 |
| FilePath | C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe | cmdline | 1 |
| FilePath | C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\AUDIODG.EXE | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\ApplicationFrameHost.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\SearchIndexer.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\SecurityHealthService.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\SgrmBroker.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\lsass.exe | cmdline | 1 |

| Type | Value | Source | Count |
|---|---|---|---|
| FilePath | C:\\Windows\\system32\\services.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\wbem\\wmiprvse.exe | cmdline | 1 |

## 10. METHODOLOGY

**Acquisition:** VBoxManage debugvm dumpvmcore — cold hypervisor-level acquisition. No kernel module or guest agent. Forensically sound. SHA256 + MD5 hash recorded on host immediately post-acquisition.

**Volatility 3:** Memory parsed using Volatility 3: pslist, pstree, cmdline, dlllist, malfind, netscan, threads. Exported as CSV (engine) and JSONL (audit trail).

**Process diff:** Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs are reassigned on every reboot and produce false-positive churn exceeding 90% of all processes.

**Risk scoring:** Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline pattern +40 · Known-malicious process name +50 · Suspicious DLL load (T1574) +30 · Anomaly name bonus +15. Thresholds: CRITICAL ≥80 HIGH ≥50 MEDIUM ≥20 LOW <20.

**Malfind filter:** Known-clean JIT processes excluded (browsers, Defender, .NET, OneDrive). Self-match false positives suppressed (e.g. lsass.exe matching itself). Only genuinely suspicious PAGE_EXECUTE_READWRITE regions retained.

**DLL analysis:** windows.dlllist CSV analysed for T1574 DLL Search Order Hijacking. Protected DLLs (amsi.dll, version.dll, cryptbase.dll etc.) checked for loads from non-System32 paths. Findings merged into ATT&CK; tagged output.

**ATT&CK; mapping:** 30+ rule-based signatures across 8 MITRE ATT&CK; tactics. Keyword matching on cmdline args, process names, DLL paths, and network artefacts. T1574-specific patterns include: copy-item, amsi.dll, invoke-atomictest.

**IOC extraction:** Regex-based extraction: IPv4 (excluding RFC1918 private ranges), domains, URLs, MD5/SHA1/SHA256 hashes, Windows file paths.

**Framework:** MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project — University of Roehampton 2026. Tool chain: VBoxManage → Volatility 3 → MFF v2 → HTML + PDF reports. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, export_alert, report_generator.

**Risk scoring:** Transparent additive model — no ML. Each evidence category contributes once. Weights calibrated to specificity (ACPO P2). CRITICAL>=80, HIGH>=55, MEDIUM>=25.

**Standards:** ACPO Good Practice Guide for Digital Evidence (Principles 1-4). NIST SP 800-86 Integrating Forensic Techniques into Incident Response. ISO/IEC 27037:2012 Guidelines for Identification, Collection and Preservation. College of Policing Digital Forensics Guidance. CPS Disclosure Manual.